Interoperability among different systems in smart city using custom blockchain.

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Problem Statement

Interoperability among different systems in smart city using custom blockchain.

Why Interoperability across domains is important?

- **Efficiency and Optimization:**
  - Interoperability allows efficient resource utilization.
- **Cost Savings:**
  - Interoperability can result in cost savings.
- **Disaster/Emergency Management:**
  - Interconnected systems provide real-time data for emergency response, enabling effective coordination in managing disasters or crises.

Use Case

Data exchange across domains in case of traffic accident

In this scenario, we have multiple domains involved, including insurance claims, police reports, and healthcare records, and we'll show how blockchain enhances interoperability.

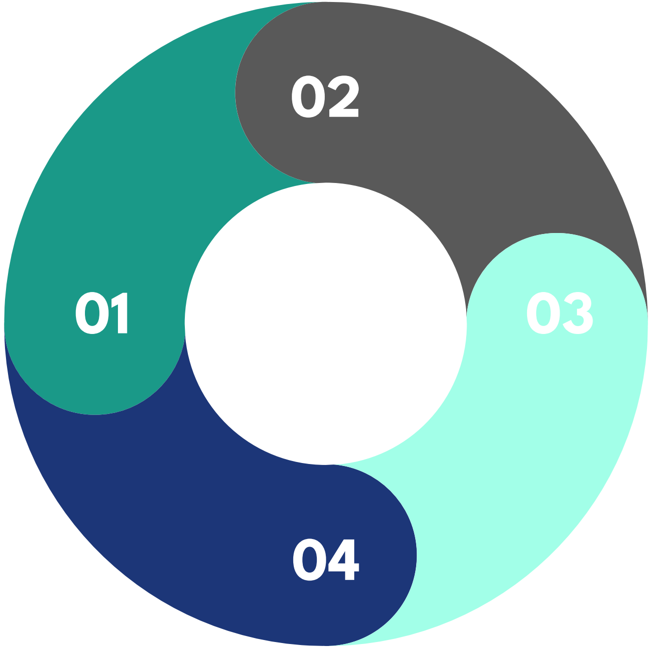
Advantages of Blockchain Interoperability for the given use case

Cross-Domain Access

Different domains can access and reference the same data on the blockchain

Security and Immutability

Data on the blockchain is secure, timestamped, and tamper-proof, ensuring the integrity of records



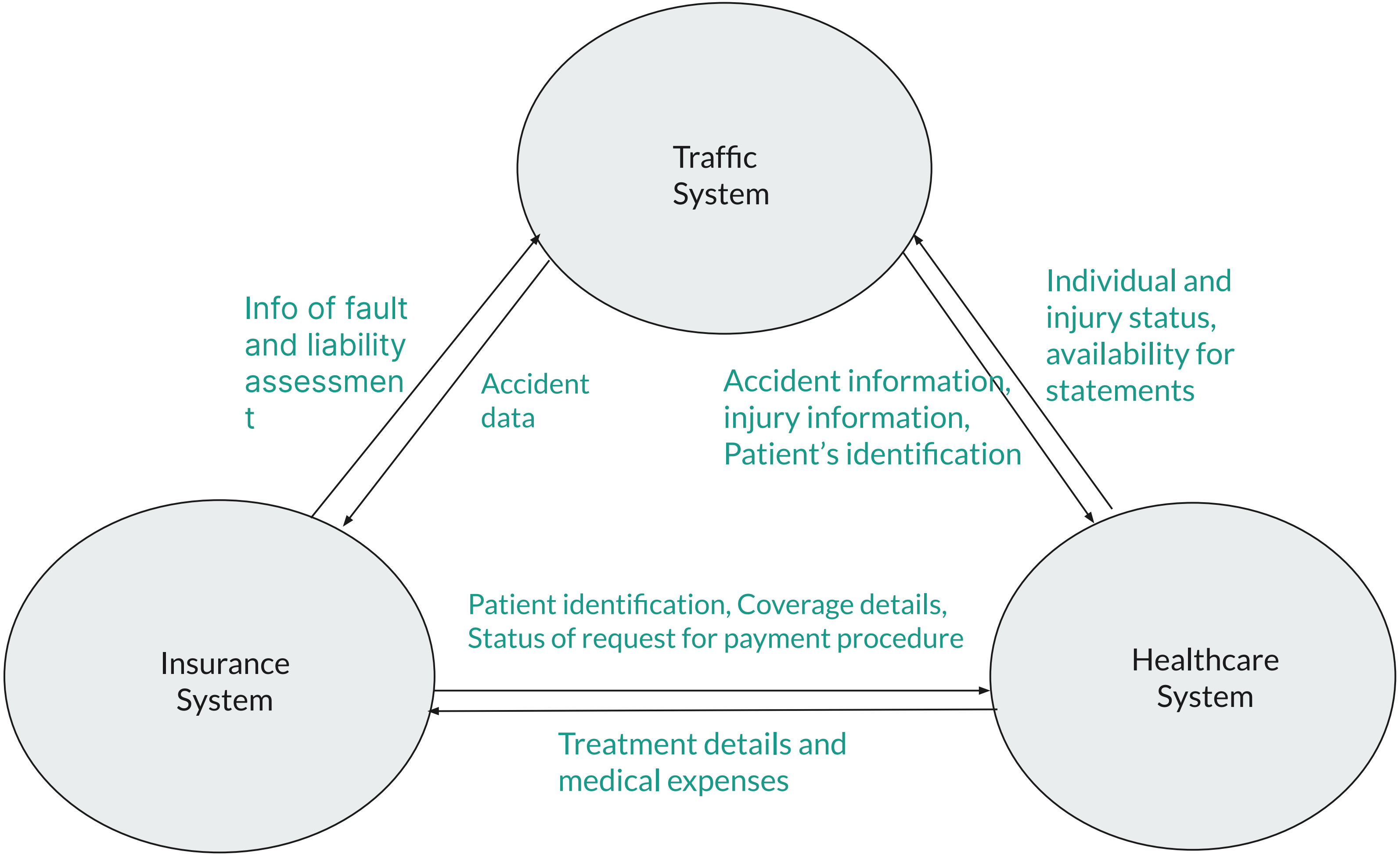
Automation

Smart contracts automate processes like claims processing, ensuring faster and more efficient interactions between domains.

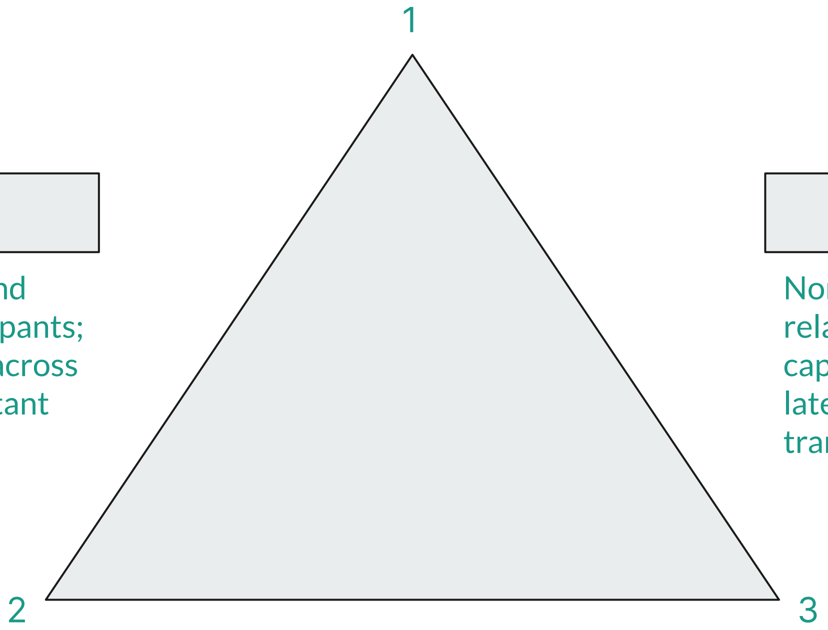
Transparency

Parties have a clear, transparent view of the entire accident-related data.

Systems in interoperability for chosen use Case



Challenges



Decentralization

Accessibility, availability and transparency for all participants; consistent state of ledger across all nodes; censorship resistant

Security

Non-functional requirements related to the ledger's carrying capacity, throughput, and latency of processing transactions

Scalability

Ability to maintain ledger integrity against malicious attacks through inbuilt control mechanisms

Scalability & Throughput

- Blockchain networks are designed to be decentralized and secure, which often results in a trade-off with scalability.
- In a traditional blockchain, every node in the network must validate and store all transactions. As more transactions are added, the network can become slower and less efficient.
- Throughput refers to the number of transactions a blockchain can process within a given time frame.
- Many blockchain networks struggle with slow confirmation times and low throughput, which can be a significant issue when dealing with real-time IoT data.

Consensus Mechanism

What is a Consensus Mechanism ?

- A consensus mechanism in a blockchain network is a **set of protocols ensuring agreement** among nodes on the state of a digital ledger.
- Serves as the verification standard for approving transactions, keeps system operational, instill trust in a trustless environment.
  Primary goal: To prevent decentralization issues like double spending.
- **“One of the most helpful examples of consensus mechanisms in action is the way that humans agree on the rules of a game”**.
- Here are the 4 most common consensus practices.
  POW - Proof of work
  POS - Proof of stake
  POA - Proof of Authority
  DPOS - Delegated Proof of Stakes.

PROOF OF WORK	PROOF OF STAKE	PROOF OF AUTHORITY
Proof of work involves miners or validators solving mathematical problems to verify transactions and earn rewards. Prons: decentralization and security Cons: slow transactions, high fees, and environmental impact. Examples: Bitcoin, Dogecoin, and Litecoin.	In a proof-of-stake model, users stake tokens to become validators, earning rewards while contributing to the network. Validators are randomly selected, with consensus breakers forfeiting their stake. Pros: scalability, energy efficiency, and cost-effectiveness. secure than proof-of-work, with power delegated by wallet size. Cons: less decentralized Examples: Ethereum, Cardano, Tezos, Algorand.	Favored by private blockchains, proof-of-authority selects validators based on reputation, not digital assets. Validators are pre-approved through vetting, often involving background checks. Pros: high scalability and minimal computing power. Cons: compromises decentralization and forfeits validator pseudo-anonymity. Examples: Xodex, JP Morgan (JPM Coin), VeChain (VET), and Ethereum Kovan testnet.

DELEGATED PROOF OF STAKE

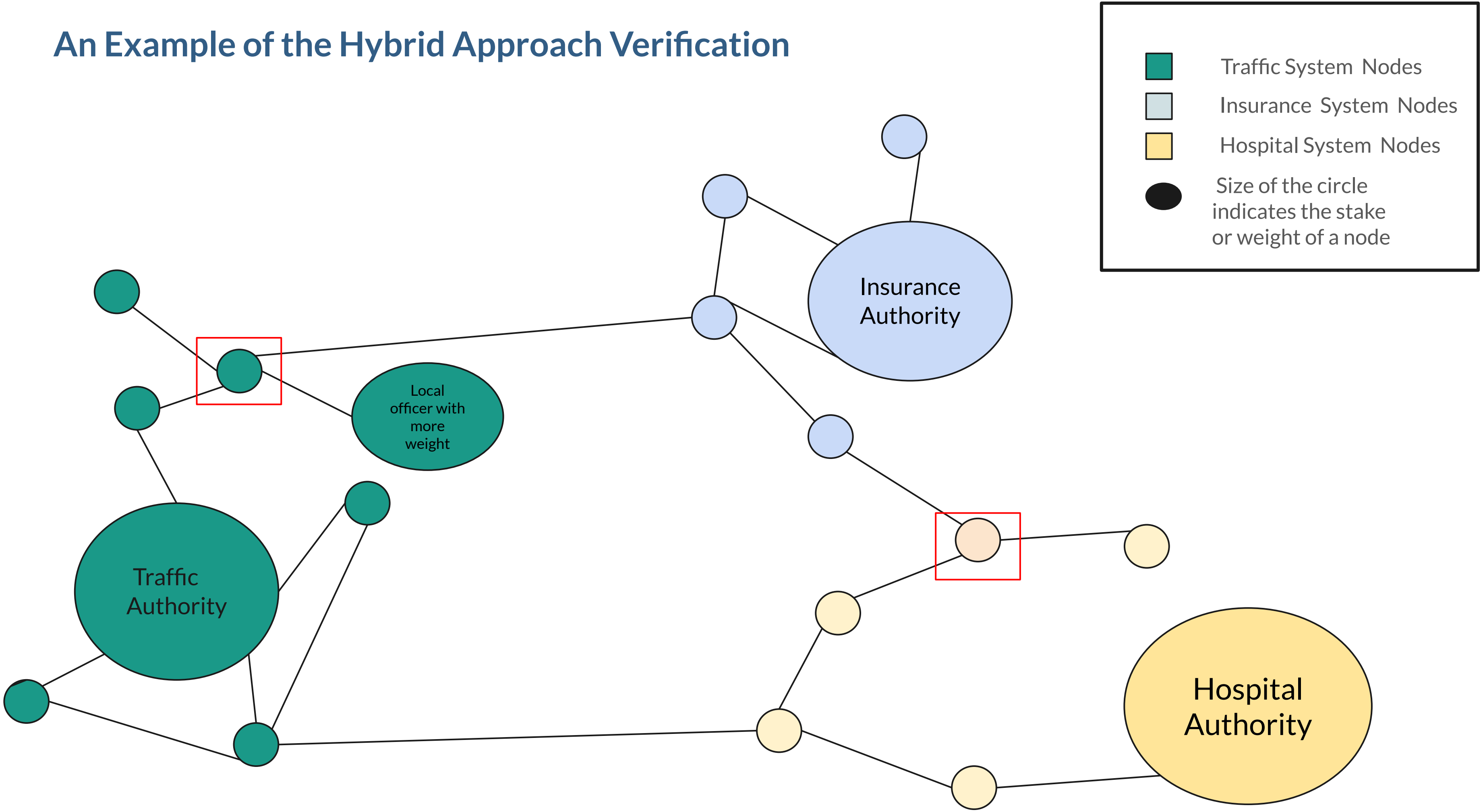
- Proof of Stake with an electoral process involves network nodes voting for delegates based on reputation, with validating privileges randomly rewarded to these delegates.
- Pros: efficiency, democracy, financial inclusivity, and accountability incentives for validators.
- Cons: tendency towards centralization, requiring high maintenance and active participation, and vulnerability to malicious actors, like in a 51 percent attack.
- Examples: EOS, Lisk, Ark, Tron, BitShares, Steem.

A Hybrid Approach

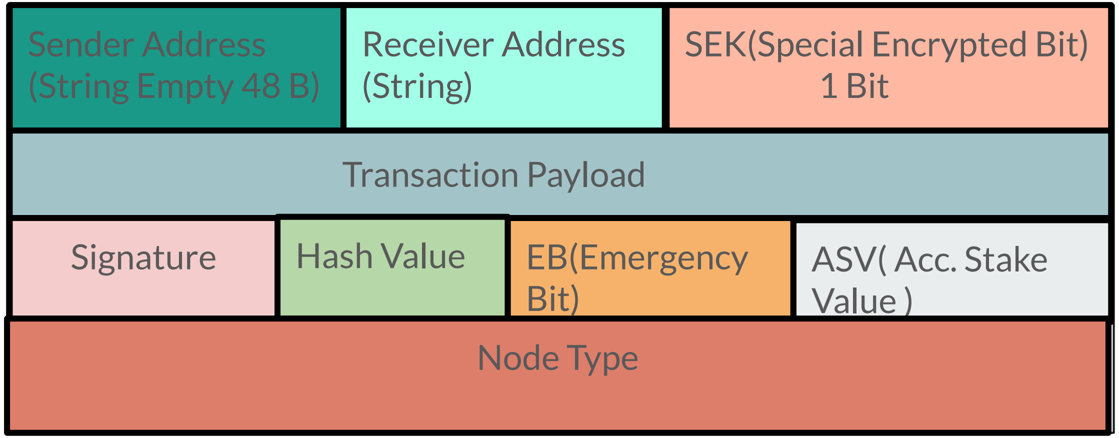
- A Hybrid approach of POS & POA is chosen to be used ,to get the gains of both the models.
  - Implemented at two level Local level of the system and Global level of the system
    - Local Level : POS (Proof of Stake)
    - Global Level : POA(Proof of Authority)
- POS can be used at a local level and POA can be used after not achieving the required threshold of stake from the local i.e in the sense of nodes that are primarily connected neighbours ,and then the Transaction can be sent to the POA Authority as he will verify it making the transaction verification successful.
- While using POS at a local level ,Nodes should verified by the node of same network.It helps to maintain the privacy of each system ,and this decision is made on an assumption that the node part of a system knows well on how to verify the transaction of that system.
- As POA is similar to a centralized approach the presence of POS at a local level will minimize the load of the Transactions on the POA authority.

Let's look into the following example for a clear view !!

An Example of the Hybrid Approach Verification

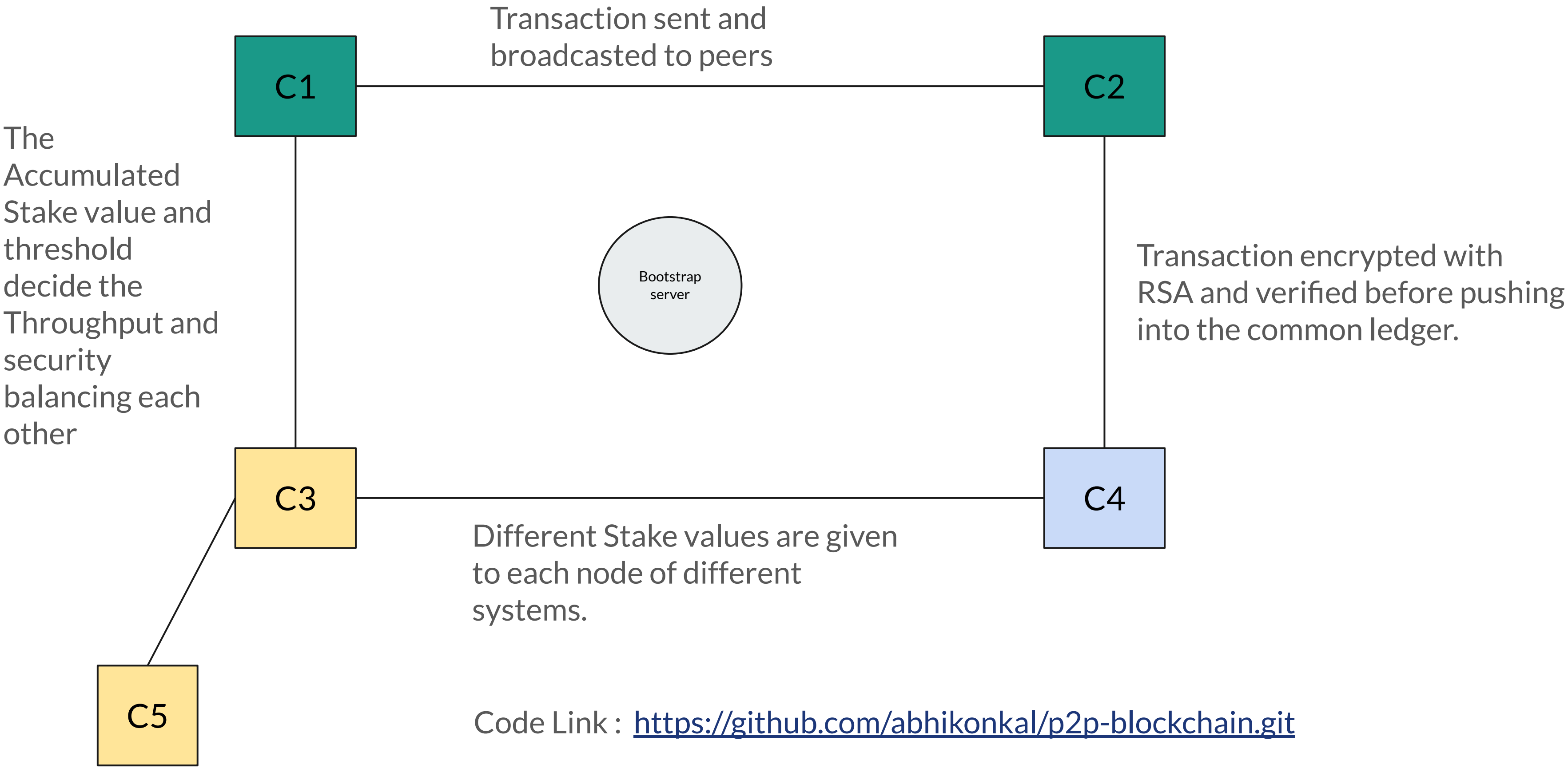


Transaction Structure



Implementation of the Idea

- So far the code has been written for a peer to peer network ,starting from 4 nodes to initiate the network .
- Transaction sent to one node are broadcasted to its neighbouring nodes.
- The peer to peer nodes of the network divided into nodes of each system ,and they communicate with each other i.e send Transactions to each other securely and verify the Authenticity and integrity of the received Transaction.
- RSA Algorithm is used for the Authenticity and sha256 is to be used for the integrity.
- First Steps are taken to implement the Consensus Mechanism ,Each nodes are given some weights and the transaction protocol is being altered with updated structure.



Code Link : https://github.com/abhikonkal/p2p-blockchain.git

Other Approaches We Are Exploring

Using DAG based decentralized system

Structure

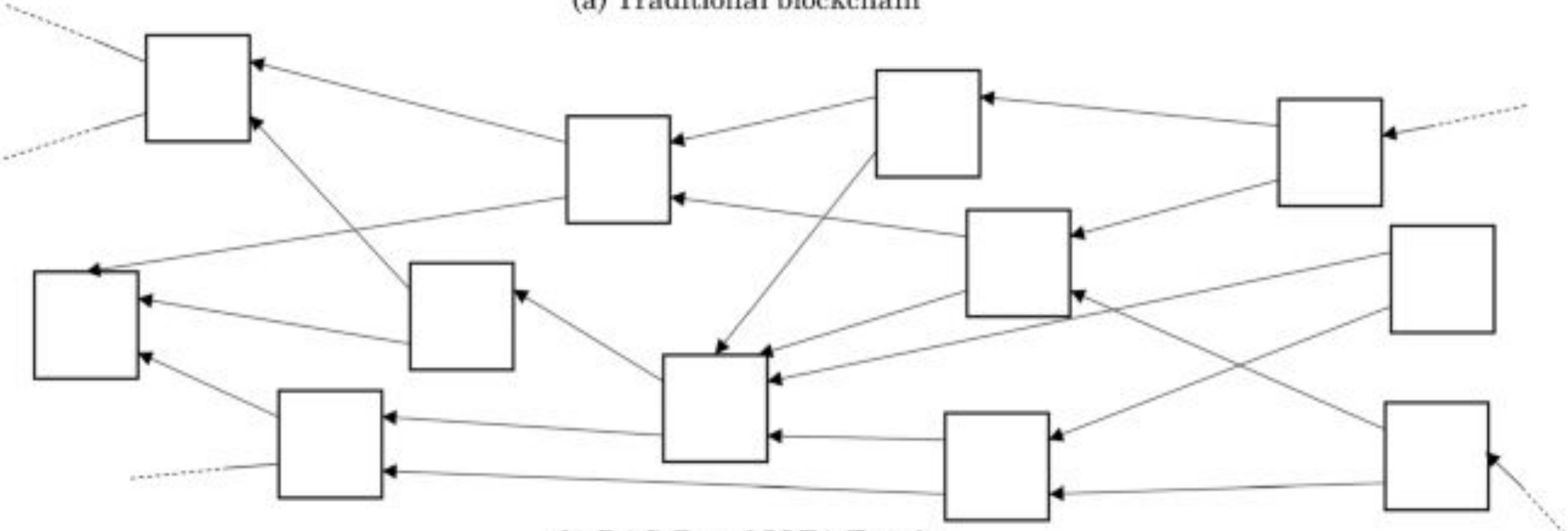
Blockchain:

- **Linear Chain:** In a traditional blockchain, transactions are organized in a linear chain of blocks. Each block points to its predecessor, forming a sequential and linear structure.

DAG:

- **Non-Linear Structure:** In a DAG, there is no fixed linear structure. Transactions are linked in a more complex graph, allowing for multiple transactions to be confirmed simultaneously, rather than waiting for a single block.

(a) Traditional blockchain



(b) DAG-Based IOTA Tangle

Confirmation and Validation

Blockchain:

- **Sequential Confirmation:** In a blockchain, transactions are confirmed sequentially in blocks. The confirmation time depends on factors like block creation time and network congestion.

DAG:

- **Parallel Confirmation:** In a DAG, multiple transactions can be confirmed simultaneously. Each new transaction references multiple previous transactions, allowing for parallel confirmation and potentially faster transaction throughput.

Consensus Mechanism

Blockchain:

- **Proof-of-Work (PoW) or Proof-of-Stake (PoS)**: Traditional blockchains typically use PoW or PoS as their consensus mechanism. Miners or validators compete to add the next block to the chain.

DAG:

- **Asynchronous Byzantine Fault Tolerance (aBFT)**: DAGs often use consensus mechanisms designed to achieve agreement among nodes without a strict, synchronized order.

Security Considerations

Blockchain:

- **Proven Security**: Traditional blockchains have been around for a longer time and have demonstrated robust security.

DAG:

- **Security Challenges**: Some DAG implementations may face challenges related to securing the network, particularly when the DAG structure and consensus mechanisms are relatively new or less tested.

Roadmap To the Future

1. Successfully Implement the Hybrid Consensus Mechanism taking help of DAG
2. Address the scalability ,Throughput vs Security concern for a large scale Data.
3. Simulating the Model for real time Data and testing the Model Blockchain for all scenarios and adapting along with new Blockchain ideas.
4. Successfully Making the Blockchain Robust and secure for Realtime.

References

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Thank You